

Prompt Engineering and Large Language Models

Midterm Examination

Weeks 1–7 · 30 questions × 2 points = 60 points, scaled to 40% of the course grade

Name: _____ Student No: _____ Date: _____

Instructions. This paper contains 30 multiple-choice questions. Select the **single best** answer for each. Each question is worth equal marks. Time allowed: 60 minutes. No notes or electronic devices.

1. Which of the following tasks is **generative**?
 - A) Ranking job applications by predicted suitability
 - B) Flagging a transaction as possibly fraudulent
 - C) Drafting a discharge letter for a patient's family
 - D) Detecting whether an image contains a tumour
2. Which statement about the term **prompt** is correct?
 - A) It refers only to the text you type
 - B) It includes attached documents and images
 - C) It excludes any instruction added by the product
 - D) It refers to the model's response
3. A model gives a confident, well-structured answer about a topic on which it has very little training material. What happens to its expressed confidence?
 - A) It falls in proportion to the thinness of the coverage
 - B) It stays essentially the same
 - C) It rises, because uncertainty makes output longer
 - D) It is reported as a percentage
4. An AI summary of a report is entirely accurate but omits the single most significant finding. Which failure signature is this?
 - A) Fabrication
 - B) Displacement
 - C) Omission
 - D) Silent failure
5. Why is the dangerous AI failure described in this course as “the fluent one”?
 - A) Fluent answers take longer to produce
 - B) Well-formatted, confident output resembles competent professional work and is trusted accordingly
 - C) Fluency indicates the model is guessing
 - D) Fluent text uses more tokens
6. ELIZA (1966) contained:

- A) A small knowledge base of medical facts
 - B) A simple neural network
 - C) Keyword matching and sentence reflection, with no knowledge or memory
 - D) A statistical model of English grammar
7. What is the “ELIZA effect”?
- A) The tendency of early chatbots to crash under load
 - B) The attribution of understanding to a system that produces plausible language
 - C) The improvement of output when a persona is specified
 - D) The degradation of models trained on their own output
8. The AI winters were primarily caused by:
- A) Technical failure of the underlying methods
 - B) The gap between what was promised and what was delivered
 - C) Shortages of computing hardware
 - D) Legal restrictions on research
9. Which best describes the constant direction of change across the four eras of the field?
- A) From general systems towards specialised ones
 - B) From open research towards commercial secrecy
 - C) From human-written rules towards learning from data
 - D) From text towards images
10. Which is **not** typically a single token?
- A) the
 - B) hospital
 - C) a rarely-used technical term
 - D) and
11. A long agglutinated Turkish word is likely to be split into several tokens because:
- A) Turkish grammar is irregular
 - B) The complete assembled word appeared rarely in training text, though its pieces are common
 - C) Turkish uses diacritics
 - D) Tokenizers cannot process non-English alphabets
12. Which is **not** a consequence of Turkish costing more tokens than English?
- A) The context window fills faster
 - B) Metered usage costs more for the same content
 - C) The model refuses Turkish input
 - D) Performance is often weaker due to thinner training coverage
13. The model’s core operation is best described as:
- A) Looking up the most relevant stored fact
 - B) Producing a probability distribution over possible next tokens

- C) Applying a decision tree written by its developers
 - D) Searching a database of documents
14. Which statement about the model's internal state is correct?
- A) It knows when it is guessing but does not report it
 - B) It performs the same operation whether the pattern is strong or weak
 - C) It has a separate mode for uncertain answers
 - D) It calculates a confidence score for each answer
15. Why can these systems not simply reply "I don't know" whenever appropriate?
- A) The phrase is filtered out by safety systems
 - B) "I don't know" is itself a token sequence competing with all others, and it is rare after questions in training text
 - C) The model is instructed never to refuse
 - D) It would use too many tokens
16. The context window contains all of the following **except**:
- A) Your current message
 - B) Documents you attached
 - C) Hidden instructions added by the product
 - D) Everything the model learned during training
17. When a document exceeds the context window, the most dangerous behaviour is:
- A) The tool refuses the input with an error
 - B) Part of the material is silently dropped and a confident answer is produced anyway
 - C) The response is truncated visibly
 - D) The model asks you to shorten the document
18. According to the four levers, which is **not** one of them?
- A) Context
 - B) Task
 - C) Format
 - D) Temperature
19. Output is accurate and complete but reads like nothing you would ever send. Which lever should you pull?
- A) Context
 - B) Task
 - C) Format
 - D) Examples
20. Which is the strongest reason to split "summarise, translate, and identify risks" into three separate turns?
- A) It uses fewer tokens overall
 - B) Each job receives fuller attention, and you can check each result before building on it
 - C) The model cannot perform three tasks

- D) Translation requires a different model
21. A prompt instructs: “You are a world-class expert with thirty years of experience and three doctorates.” What does this most likely achieve?
- A) A measurable improvement in factual accuracy
 - B) A shift in register and vocabulary, with the superlatives adding little
 - C) Access to specialist training data
 - D) A reduction in hallucination
22. Someone adds a phrase to a prompt, the output improves, and they conclude the phrase works. The strongest objection is:
- A) The phrase may be too long
 - B) Output varies between runs, so the improvement may be unrelated to the change
 - C) The model may not have read the phrase
 - D) The phrase may be in the wrong language
23. Which instruction converts an unverifiable claim into a checkable one?
- A) “Be accurate and thorough”
 - B) “Only use reliable sources”
 - C) “After each claim, quote the sentence that supports it”
 - D) “Think carefully before answering”
24. In the three dimensions of register, **field** refers to:
- A) The subject area and its vocabulary density
 - B) The relationship between writer and reader
 - C) The channel of communication
 - D) The emotional tone
25. Which specification is most likely to produce reproducible style?
- A) “Make it professional and friendly”
 - B) “Everyday vocabulary with no unglossed medical terms; warm but not casual; written to be read aloud”
 - C) “Use an appropriate tone for the audience”
 - D) “Write it well”
26. Why does a stated fallback such as “write NOT STATED” reduce invented content?
- A) It reduces the model’s temperature
 - B) It makes the honest answer a probable continuation rather than an unlikely one
 - C) It prevents the model from accessing training data
 - D) It shortens the output
27. Which is the most effective way to specify institutional house style?
- A) A list of adjectives describing the desired tone
 - B) Two or three short samples of approved writing
 - C) An instruction to write “in our usual style”
 - D) A request for formal language

- 28.** A generated browser tool produces a subtly wrong calculation but has a clean, working interface. Why is this particularly dangerous?
- A) The error is intermittent and therefore hard to reproduce
 - B) The error is consistent, which makes the tool appear reliable
 - C) The tool will eventually stop working
 - D) The interface will change without warning
- 29.** Grounding a system in your own documents changes:
- A) What the model knows
 - B) Where the answer comes from
 - C) Whether the model checks the truth of its output
 - D) The model's training data
- 30.** An assistant grounded in your documents produces an answer with no supporting quotation. The most likely explanation is:
- A) It over-read the passage
 - B) It retrieved the wrong passage
 - C) It answered from memory
 - D) The context window overflowed